# Expose and Conquer: Fully Test-time Adaptation via Unlearning Nuisance Factors

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## ABSTRACT

Deep neural networks suffer a steep drop in performance when facing the domain shift problem. A typical strategy, which involves aligning the source and target feature distributions, becomes infeasible during fully test-time adaptation (FTTA) due to the lack of access to source data. Addressing this issue, our work introduces a novel expose-and-conquer strategy to foster shift-invariant representation within FTTA constraints. We redefine unwanted shifts between the source and target domains as nuisances and mimic these nuisances in the target data with augmentations. During adaptation, we enforce the source model to unlearn the nuisances by removing their effects on the unwanted feature shifts while maintaining confident and consistent label predictions. To systematically eliminate the influence of nuisances, we devise a novel mutual information-based criterion that operates within the feature space. Facilitating shift-invariant feature learning at test time, our method enhances the source model's generalization capabilities to the target domain under FTTA. Extensive experiments on various tasks, involving corruption shifts and style shifts, demonstrate our method can consistently achieve outstanding performance and reach state-of-the-art on most benchmarks.

## CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; **Learning under covariate shift**; *Learning latent representations*; *Kernel methods*.

## KEYWORDS

Test-Time Adaptation, Invariant Feature Learning, Disentangle Nuisance Factors

## 1 INTRODUCTION

Deep neural networks (DNNs) have demonstrated excellent performance across numerous applications when test and training data are independently and identically distributed [10, 25, 27, 42, 43]. However, under domain shifts, where the training (source) data differ from the test (target) data, DNN models suffer a precipitous drop in accuracy [20]. Unlike the human visual system which is robust to shifts in images, DNNs struggle with classification when the test data undergo even simple corruptions or general style changes, such as rotation or blur [14, 20, 34], or synthetic-to-real [35] shift.

To cope with such domain shift problems that naturally occur in real-world applications, domain adaptation techniques have been widely studied [11–13, 31, 51]. Prevailing domain adaptation approaches mostly involve minimizing the divergence between the source and target feature distributions, encouraging the source model to learn domain invariant representation and improve its generalizabilityility on the target data [4, 37, 38, 44, 47]. These adaptation strategies rely on the accessibility of labelled source data, and target data, which is hard to satisfy due to privacy issues or labour-intensive annotations in practical tasks. Test-time training (TTT) [30, 46] is proposed to ease the requirement for annotated target data via self-supervised learning (SSL), but it requires the source model to be re-trained jointly on the auxiliary SSL task. Some works explore source-free domain adaptation with additional generative models to support feature alignment [26, 28, 53]. However, it requires significant modifications to the overall model backbone and training procedure to employ these adaptation techniques.

In this paper, we focus on the challenging but practical setting of Fully Test-Time Adaptation (FTTA), where no target domain annotation, no source data and no specialized training protocol of the source model are required. We seek to learn shift-invariant feature representations at test time without accessing source data and source model re-training, while merely adapting our model weights with the unlabelled data from the target domain.

Our question is: *How can a pre-trained source model be adapted to fit the target dataset with domain-invariant features when its training data is inaccessible?* To address this challenge, we specifically focus on the data shift between the source and target domains and adopt a novel expose-and-conquer strategy. We argue that if the adapted model is robust to the domain shift between the source and target, then it should also demonstrate robustness to the exactly mimicked shift, or even to amplified shifts, applied to the target domain. At test time, without prior knowledge of the shift between the source and target domain, we explicitly mimic undesired data shifts between the source and target domain with common augmentation techniques and apply them to the test set. These exemplified test data shifts involve nuisance factors *w.r.t.* the main classification task into FTTA, where the adapted model should be encouraged to unlearn the effects of these nuisances and seek feature invariance.

To facilitate nuisance factor unlearning during adaptation, we specifically enforce the source model to remove the test feature shift caused by the nuisances. We propose to minimize the mutual information between the extracted data feature and the feature shift introduced by the nuisances, and present a a matrix-based Renyi's entropy functional algorithm to estimate the term efficiently. With the proposed criterion, disentangling and removing the influences of nuisances at test-time feature learning, the adapted model learns shift-invariant test features that benefit its generalization to the target domain without requiring source data and source model training protocols. Additionally, we regularize the model to achieve confident and consistent label predictions on the original test data and nuisance-shifted data, to further enhance shift-invariance of the extracted test features. Collectively, our proposed adaptation method, named **T**est-time **I**nvariant **R**epresentation learning through **N**uisances **U**nlearning (TIRNU), leads to shift-invariant and compact test data representations that are independent of the nuisances that mimic the source-target shifts. This can consequently increase the model generalization ability to the target domain.

Our work conveys a key message: mimicking nuisance-augmented data that represents unwanted shifts between the source and target domains can help address practical domain shift challenges when source data is not accessible. Our proposed TIRNU approach enables shift-invariant representation learning without accessing the source data. It mimics human learning, where we can disregard nuisance factors that are uninformative to task-dependent feature extraction while remaining confident with our predictions when adapted to transformed images. Our proposed nuisance unlearning strategy analogizes the human cognitive function of stimulus strengthening and active forgetting [17]. Extensive experiments verify TIRNU can achieve outstanding performance in various adaption tasks, involving different domain shifts. Moreover, TIRNU opens up the flexibility to benefit domain adaptation with arbitrary prior knowledge on the shift between the source and target domains. That is, to provide shift-specific nuisance augmentations, enhancing the model accuracy with to-the-point domain invariant features.

Our contributions are summarized as follows:

- We provide a novel perspective to achieve test-time domain invariant feature learning via nuisance-shift unlearning.
- We derive a novel mutual information-based criteria to remove the effects of the nuisances in test-time invariant feature learning. We also present additional regularizers to further enhance feature invariance under FTTA.
- Our proposed method performs competitively with other strong baselines for test-time adaptation. Across all tasks, TIRNU consistently achieves state-of-the-art results on the CIFAR10-C/100-C/10.1, Imagenet-A/Imagenet-R, and VisDA-2017 benchmark datasets.

## 2 RELATED WORK

**Test-Time Adaptation.** Test-Time Adaptation (TTA) is a special setting of domain adaptation, where we can only access the test data and a pre-trained source model to generalize a model to a target domain [48, 57]. Different from the Test-Time Training setting [30, 46] , which also works without access to source domain data, TTA further eases the requirement of source model retraining. Prior works have leveraged labelled test datasets to facilitate TTA, fine-tuning the source model to fit one-shot or few-shot test data samples [9, 57]. However, such solutions pose additional requirements for test data annotations. Our work focuses on exploring a more challenging and practical setting of fully test-time adaptation (FTTA), where annotations on test data are not available.

In Fully Test-Time Adaptation (FTTA), a key focus is defining a proper objective to optimize. Test entropy minimization (Tent) [48] addresses distribution shifts under FTTA using an entropy regularization, enforcing the model to achieve confident label prediction on test data samples. While TENT allows both online and offline updates for FTTA, Source Hypothesis Transfer (SHOT) [29] focus on exploring the offline FTTA issue of source-free adaptation with pseudo-test data labelling. It employs a clustering objective that strongly aligns with the main classification task to facilitate FTTA. However, the additional pseudo-labelling process can propagate errors, especially in the presence of a domain shift, which would misguide the model adaptation. Another aspect to consider for FTTA is what and how the parameters of the source model are to be updated. [39] suggested that if the difference between the source and target domains primarily lies with the first and second-order moments of the features, then the problem can simply be mitigated by normalization. Related works normalize test data with the test statistics as a weak form of domain alignment [6, 23, 37, 55]. Tent [48] leverages a similar idea and uses test statistics and the test entropy to update only the batch normalization layers [48]. This restriction on the parameters can result in limited adaptation, causing inferior model performance on more challenging adaptation tasks [7, 30]. SHOT [29] incorporates flexible source model parameter updates by dividing the source model into two modules: a feature encoder and a feature classifier. During adaptation, it freezes the classification head and updates the entire feature encoder, allowing the source model to sufficiently adapt to the target data while leveraging the well-trained feature classifier incorporated within the source model. In recent years, there has been emerging interest in exploring online updates for FTTA. For instance, AdaNPC [59] introduces a non-parametric classifier to facilitate the online adaption of streaming data by leveraging a historical memory bank. LAME [5] focuses on parameter-free online TTA, adjusting classifier output probabilities rather than feature extractor parameters. ENT [16] proposes conjugate pseudo-labels for online adaptation. ETA [32] achieves efficient online TTA through sample-efficient entropy minimization. MEMO [57] emphasizes one-shot updates and adapts models using entropy from marginal output distribution over augmentations. SAR [33] specifically explores stabilizing online TTA using a sharpness-aware learning scheme.

Our approach is general to work under both the offline and online settings of TTA. We are original in our motivation to leverage prior knowledge on the potential shift between source-and-target domain to benefit test-time-adaptation. Our approach diverges from prior methods in our explicit aim of seeking domain-invariant representation with only the test dataset. Although learning domain-invariant features through source-target feature alignment has been widely studied in prior works [26, 28, 37, 38, 51, 53], to the best of our knowledge, we are the first to explore the specific strategy of learning without nuisance, i.e. involving nuisance factors and explicitly unlearning their influences, to facilitate fully test-time adaptation.

**Nuisance Factors Disentanglement.** A nuisance factor is any random variable that affects the observed data but is not informative to the task to solve [1]. In a classification task, random data corruptions or style changes that commonly trigger dataset shifts through data poisoning [24, 45] or translation [40, 49], introduce nuisances $\mathbf{n}$ that shifts the input image $\mathbf{x}$ yet is uninformative for predicting the label $y$. Some prior works have particularly leveraged the property of nuisances to benefit their objective tasks with invariant features. Theoretically, Alemi et al. proposed a feature learning algorithm that empirically shows improved resistance to adversarial nuisances [2]. Achille et al. explored feature insensitivity and invariance to nuisances of the Variational Information bottleneck under a task-nuisance decomposition assumption [1]. Application-wise, Yao et al. and Zheng et al. explored nuisance disentanglement in deep clustering to learn feature partitioning that disregards specific nuisance factors [52, 56]. Similarly, Wei et al. presented a general approach to improve model robustness with

nuisance-label supervision on top of the main task [50]. These methods require the nuisance factor to be predefined in an explicit form of labels or attributes, which brings in extra annotation burdens.

In this work, we eliminate the explicit definition of nuisance factors, and directly exploit the influence of nuisances instead. We collectively treat unwanted test data shifts performed with augmentations as nuisances and seek to unlearn the influence of nuisances by removing its triggered feature shift in the feature space.

## 3 METHOD

In this paper, we treat unwanted shifts between the source and target domain as nuisance factors and seek to unlearn the influences of nuisance at test-time, to facilitate source-free shift-invariant representation learning under FTTA. In this section, we first elaborate on our proposed TIRNU approach, including its problem statement and the model design. Then, we present the optimization algorithm of our objective, facilitated with a matrix-based Renyi's entropy functional for mutual information estimation.

### 3.1 Fully Test-time Adaptation (FTTA)

**Problem statement**: In fully test-time adaptation, given only the unlabelled test data $\{\mathbf{x}_t^i\}_{i=1}^{N_t}$ from the target domain $t$, and a source model $f_s : \mathcal{X}_s \to \mathcal{Y}_s$ pre-trained on labelled data $\{\mathbf{x}_s^i, y_s^i\}_{i=1}^{N_s}$ from the source domain $s$, we aim to learn an adapted model $f_t : \mathcal{X}_t \to \mathcal{Y}_t$ from $f_s$, such that it can infer $\{y_t^i\}_{i=1}^{N_t}$ from $\{\mathbf{x}_t^i\}_{i=1}^{N_t}$ with high accuracy. For notation simplicity, we drop the domain and index indication of the test data henceforth and use $\mathbf{x}$ to represent arbitrary test data from the target domain unless otherwise specified.

Without loss of generality, we assume the DNN-parameterized source model $f_s$ consists of two modules: the feature extractor $g_s : \mathcal{X}_s \to \mathcal{Z}_s$ where $\mathbf{z} \in \mathbb{R}^d$ and the classification module $h_s : \mathcal{Z}_s \to \mathcal{Y}_s$ where $y \in \mathbb{R}^K$ ($d$ and $K$ indicate the feature dimension and number of classes, respectively). Therefore, $f_s(\mathbf{x}) = h_s(g_s(\mathbf{x}))$. Similarly, for the adapted model on the target domain, $f_t(\mathbf{x}) = h_t(g_t(\mathbf{x}))$.

During test-time adaption, we freeze the classification module, *i.e.* $h_t = h_s$, and seek to learn an adapted feature extractor $g_t^\theta$, parameterized by $\theta$, that can learn shift-invariant features for the target test data without accessing the source data, and without modifying the training protocol of $f_s$.

### 3.2 Nuisances Augmented Data

To address the problem, we explicitly consider the shift between the source and the target domain. In our proposed TIRNU (depicted in Figure1), we treat unwanted shifts between $s$ and $t$ as nuisances that are uninformative to the main classification task. We mimic those specific nuisances onto test data samples with data augmentation. Then, for the sake of invariant feature learning, we enforce the model to unlearn the influence of nuisances at test time by removing the test feature shift while maintaining confident and consistent label predictions.

To be specific, in a general case where no prior knowledge of the source-target shift is given, we collectively mimic unwanted shifts between the source and target domain with common data augmentation strategies, such as random cropping or transformation used in simCLR [8], $L_p$ data corruptions [41], and AugMix [21]. A model with general transferability should be invariant to those common augmentations. While in scenarios where hints of potential shifts between the source and target are present, tailor-designed transformation or generation modules can be leveraged to mimic nuisance shifts for test data augmentation. These performed augmentations involve nuisance factors that would derange the source model from learning informative features for the target domain data, but should not trigger differences in the final label prediction for given test samples. This property enables us to perform source-free invariant feature learning by involving those nuisance shifts to the test data and subsequently unlearning the influences of nuisances during the test-time feature extraction phase.

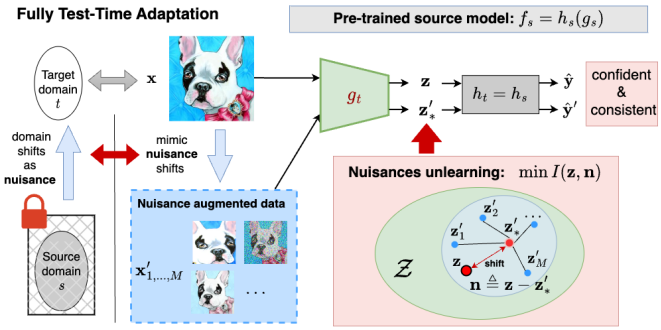


**Figure 1: Our proposed TIRNU approach for fully test-time domain adaptation. Without access to the (source) training data, we adopt an expose-and-conquer strategy: we define potential unwanted shifts between the source and target as nuisances and mimic these nuisances to the target data with augmentations. During test-time adaption, we update the feature extractor to unlearn the specific nuisance shift provided in test data augmentation, by removing their effects on the unwanted feature shifts while maintaining confident and consistent label predictions. Our model enhances the source model's generalization capabilities to the target domain with invariant feature representations.**

Suppose that these augmentation strategies define a collective set of mapping functions, which is denoted as $\mathcal{A}$, conveying unwanted domain or data shifts. Given a test sample $\mathbf{x}$, we uniformly sample $M$ transformations $a \sim \mathcal{U}(\mathcal{A})$ and obtain $M$ nuisance-shifted versions of $\mathbf{x}$, denoted as $\{\mathbf{x}'_j\}_{j=1}^M$, where $\mathbf{x}'_j = a_j(\mathbf{x})$. We explicitly define the influence of the nuisances shift on $\mathbf{x}$ as the collective feature shifts caused by those unwanted data transformations, which is denoted as

$$\begin{aligned} \mathbf{n} &\triangleq \mathbb{E}_{\mathcal{U}(\mathcal{A})}\left[g_t^\theta(\mathbf{x}) - g_t^\theta\left(a(\mathbf{x})\right)\right] \\ &\approx g_t^\theta(\mathbf{x}) - \frac{1}{M}\sum_{j=1}^{M} g_t^\theta\left(\mathbf{x}'_j\right) = \mathbf{z} - \mathbf{z}'_*, \end{aligned} \tag{1}$$

where $\mathbf{z}'_* = \frac{1}{M}\sum_{j=1}^M g_t^\theta\left(\mathbf{x}'_j\right)$ is the centroid of the shifted features $\mathbf{z}'_j = g_t^\theta\left(\mathbf{x}'_j\right)$ for $j = 1, \dots, M$. This newly introduced variable $\mathbf{n}$ captures the overall feature shift caused by the nuisance. The adapted model should unlearn that negative influence to achieve shift-invariant test features for model generalization.

### 3.3 Nuisances Unlearning

Then, to unlearn the influence of nuisances at test time, we propose to minimize the nuisance unlearning loss $\mathcal{L}_{NU}$, which is the mutual information between the extracted test data features and the collective feature shift, *i.e.*,

$$\mathcal{L}_{NU} = I(g_t^\theta(\mathbf{x}), \mathbf{n}). \tag{2}$$

This criterion forces the model to disentangle and remove the influence of nuisances when extracting informative test features for domain invariance. Through this loss, the model is adapted to be robust to the specified shift defined in $\mathcal{A}$. Therefore, in special cases where prior knowledge on the shift between the source and target is given, we can tailor-design the transformation set $\mathcal{A}$ to enable the model to achieve feature invariance *w.r.t.* specific domain shift. For instance, suppose that we know *a priori* the test data is shifted from the source data with image corruptions, $\mathcal{A}$ can be intentionally set to be common corruption algorithms, so as to enforce the adapted model to be specifically robust to data corruptions. We examine the effects of applying different schemes of $\mathcal{A}$ in Section 4.4.

Moreover, to enhance the shift-invariance of the extracted test features, we further encourage the adapted model to maintain confident and consistent label predictions on the nuisance-shifted test data. Specifically, let $\hat{y}$ denote the label prediction of $\mathbf{x}$ with $f_t$, *i.e.*, $\hat{\mathbf{y}} = h_t(z)$. Similarly, for the nuisance-shifted data $\mathbf{x}'$, we have $\hat{\mathbf{y}}' = h_t(z')$. We enforce the adapted model to achieve confident label prediction by minimizing the entropy of each prediction. We further constrain the model to maintain consistent label predictions on the original test feature and the nuisance-shifted features with cross-entropy loss terms. The individual influence of label confidence and label consistency loss terms are examined in model ablation presented in section 4.4. Collectively, our additional regularizer for feature invariance regarding label prediction is

$$\mathcal{L}_{label} = \sum_{\mathbf{z},\mathbf{z}'} H(\hat{y}) + H(\hat{y}') + CE(\hat{y}, \hat{y}') + CE(\hat{y}', \hat{y}). \tag{3}$$

where $H(\hat{y}) = -\sum_c p(\hat{y}_c) \log p(\hat{y}_c)$, $H(\hat{y}') = -\sum_c p(\hat{y}'_c) \log p(\hat{y}'_c)$, $CE(\hat{y}, \hat{y}') = -\sum_c p(\hat{y}_c) \log p(\hat{y}'_c)$ and $CE(\hat{y}', \hat{y}) = -\sum_c p(\hat{y}'_c) \log p(\hat{y}_c)$. For all terms, $p$ denotes the predicted probability.

Therefore, the overall objective of our TIRNU approach is

$$\min_\theta \ \mathcal{L}_{NU} + \lambda \mathcal{L}_{label} \tag{4}$$

where $\lambda > 0$ is a hyperparameter balancing the loss term of explicit nuisance unlearning in the feature space and the regularization term for confident and consistent final label prediction.

### 3.4 Optimization

The optimization of our objective is initially challenging, since our proposed information-based nuisance unlearning loss $\mathcal{L}_{NU} = I(g_t^\theta(\mathbf{x}), \mathbf{n})$ in Eq.(2) is computationally intractable. To tackle the issue, we present a matrix-based Rényi's $\alpha$-order entropy functional to estimate this mutual-information-based loss term, allowing a direct and efficient optimization of our proposed TIRNU.

As proved in [54], the matrix-based entropy functionals provide a powerful tool to estimate variable dependencies without explicit estimation of the underlying data distributions, while in a differentiable manner. Compared with mutual information estimators that resort to variational approximation or adversarial training, such as MINE [3], a matrix-based estimator is mathematically well-defined and computationally more efficient, especially for large networks [58]. Leveraging a differentiable matrix-based mutual information estimator for the $\mathcal{L}_{NU}$ term in our objective, we can directly update TIRNU with a uniform gradient-based optimizer.

To be specific, our proposed nuisance unlearning criterion $\mathcal{L}_{NU} = I(g_t^\theta(\mathbf{x}), \mathbf{n}) = I(\mathbf{z}, \mathbf{n})$ can be rewritten as

$$I(\mathbf{z}, \mathbf{n}) = H(\mathbf{z}) + H(\mathbf{n}) - H(\mathbf{z}, \mathbf{n}) \tag{5}$$

Inspired by [58], we estimate each entropy term $H$ using the matrix-based Rényi's $\alpha$-order entropy functionals, $H_\alpha$, which is precisely Shannon's entropy when $\alpha \to 1$. Specifically, given a mini-batch of $B$ data samples, we obtain both random vectors $\mathbf{z}$ and $\mathbf{n}$ from the adapted feature extractor $g_t^\theta$. According to [15], the entropy of $z$ can be defined over the eigenspectrum of a normalized Gram matrix $A_\mathbf{z} = K_\mathbf{z}/\mathrm{tr}(K_\mathbf{z}) \in \mathbb{R}^{B\times B}$, where $K_\mathbf{z}(i,j) = \kappa(\mathbf{z}_i, \mathbf{z}_j)$ and $\kappa$ is the Gaussian kernel. That is,

$$H_\alpha(A_\mathbf{z}) = \frac{1}{1-\alpha} \log_2 \mathrm{tr}\,(A_\mathbf{z}^\alpha) = \frac{1}{1-\alpha} \log_2 \left( \sum_{j=1}^{n} \lambda_j (A_\mathbf{z})^\alpha \right), \tag{6}$$

where $\lambda_j$ is the $j$-th eigenvalue of $A_\mathbf{z}$. The entropy of $\mathbf{n}$ can be similarly defined based on another normalized Gram matrix $A_\mathbf{n} \in \mathbb{R}^{B\times B}$. Further, the joint entropy for $\mathbf{z}$ and $\mathbf{n}$ can be defined as

$$H_\alpha(\mathbf{z}, \mathbf{n}) = H_\alpha \left( \frac{A_\mathbf{z} \circ A_\mathbf{n}}{\mathrm{tr}(A_\mathbf{z} \circ A_\mathbf{n})} \right), \tag{7}$$

where $\circ$ denotes the Hadamard product. With Eq.(5)-(7), the matrix-based Rényi's $\alpha$-order mutual information estimation $I_\alpha(\mathbf{z}, \mathbf{n})$ in analogy of Shannon's mutual information is given by:

$$I_\alpha(\mathbf{z}, \mathbf{n}) = H_\alpha(A_\mathbf{z}) + H_\alpha(A_\mathbf{n}) - H_\alpha(A_\mathbf{z}, A_\mathbf{n}) \tag{8}$$

which we estimate $I(\mathbf{z}, \mathbf{n})$ with $I_\alpha$ by setting a value of $\alpha$ close to 1. Note that, we empirically testified that our model performance is not sensitive to this $\alpha$ hyperparameter (see Figure 2). Algorithm 1 presents the overall scheme of our TIRNU approach for offline adaptation. For online adaptation, Algorithm 1 continues iterating over new batches of incoming data.

**Dynamic Queue** Empirically, for our TIRNU objective, all entropy-based loss terms are to be computed over a batch of data. In our implementation, to eliminate the potential bias on more recent mini-batches when updated with small batch sizes, we maintain a dynamic queue of several mini-batches to compute those mutual information and entropy losses [18, 30]. By maintaining a dynamic queue, our estimated mutual information and entropy losses are more representative of the target domain and would decouple the model dependence on the batch size (examined in section 4.4).

**Algorithm 1** Test-time Invariant Representation learning through Nuisances Unlearning (TIRNU)

**Require:** Source pre-trained model $f_s$ (adapt $g_s$ to get $g_t$, fix $h_s$), test data $\mathcal{X}_t = \{(\mathbf{x}_t^i)\}_{i=1}^{N_t}$, set of augmentations $\mathcal{A}$.

```
for epoch= 1 to E do
    for minibatch= 1 to Q do
        X ← SampleMiniBatch(X_t)
        Get augmented inputs X' ← a(X), where a ~ U(A)
        Z, Z' ← Forward (X, X', g_s)
        Obtain nuisance N from Z, Z'
        Estimate L_NU with Z, N using Equations 5, 6, 7
        Y, Y' ← Forward (X, X', h_s)
        Compute L_label with Y, Y' using Equation 3
        Update g_s via SGD with objective L_NU + λ L_label
    end for
end for
```

## 4 EXPERIMENTS

**Datasets** To assess our method's robustness against real-world image distortions across diverse domains, we conduct evaluations in three key settings: mixed image corruptions, natural adversarial examples, and style transfer scenarios. These three scenarios cover a wide range of domain shifts studied in the literature. Our experiments are conducted on six different distribution shift benchmark datasets: CIFAR10-C/CIFAR100-C [20], CIFAR10.1 [36], Imagenet-A [22], Imagenet-R [19], and VisDA-2017 [35]. To be more specific,

- The **CIFAR10-C** and **CIFAR100-C** consist of noisy images involving different types of corruption. We test TIRNU on those two datasets to assess its robustness to common distributional shifts in images.
- The **CIFAR10.1** and **Imagenet-A** datasets include challenging adversarial examples that are visually not too dissimilar to the source domain. CIFAR10.1 data are sampled from the TinyImagenet, while Imagenet-A consists of adversarial samples that have been wrongly classified by ResNet models. Together, CIFAR10.1 and Imagenet-A could be considered to contain natural adversarial samples that are straightforward to identify by the human eye, but difficult for neural networks to classify.
- **Imagenet-R** contains a collection of renditions of 200 classes of Imagenet images, and the **VisDA-2017** dataset is a large synthetic-to-real benchmark with 12 classes. We conduct experiments on those two datasets to demonstrate how our approach performs when the target test data are from a different style compared to the source training data.

**Baselines** Our TIRNU allows both offline and online updates for adaptation. Here, we compare our approach to prior test-time adaptation methods under both problem settings. We compare with TestBN, SHOT, TENT, and TT-C under the offline source-free adaption setting; and compare with MEMO and SAR under the online setting for model updates with streaming data.

- **Source**: the pre-trained source model that is evaluated on the test data without any adaptation.
- Test-time batch normalization (**Test BN**) [39]: estimate the test data statistics and use it to normalize the BN layers to update the source model.
- Source hypothesis transfer (**SHOT**) [29]: on top of test entropy, SHOT minimizes a global diversity term and a cross-entropy term utilizing pseudo-labels, and updates the source feature extractor.
- Test entropy minimization (**Tent**) [48]: Tent minimizes the test entropy, and updates only the affine parameters of the BN layers in the source model.
- Test-time contrastive learning (**TT-C**) [30]: an FTTA ablated version TTT++. While the TTT++ requires information on the source data, we adopt TT-C to merely minimise an auxiliary contrastive loss at test time and update the source feature extractor and the self-supervised head. Note that, we highlight the distinction of TT-C to the TTT-C model presented in the paper: TTT-C operates on a model that was jointly trained on the main classification task and an auxiliary SSL task, whereas for TT-C, the model was simply trained on the main task, which is more appropriate for the fully test-time setting.
- Marginal entropy minimization with one test point (**MEMO**) [57]: An online FTTA method that performs one-shot updates and adapts models using entropy from marginal output distribution over augmentations of the test data.
- Sharpness-aware and reliable entropy minimization method **SAR** [33]: an online FTTA method that performs stabilized updates with a tailor-designed optimization scheme.

SHOT and Tent are the most popular methods in offline fully test-time adaptation. TTT++, the full model of TT-C, reported much greater gains in performance compared to SHOT and Tent, but at the cost of additional source model training protocol, and the requirement for the source data statistics. Since TTT++ demonstrated superior accuracy to both SHOT and Tent, we include TT-C as an additional baseline to examine how it performs in the fully test-time case. We present additional experiment comparisons with MEMO and SAR to demonstrate the effectiveness of TIRNU under the online-update setting of adaptation.

**Implementation Details** For all the datasets, the source model given is a ResNet-50 pre-trained on the source training data using the standard training procedure. For example, for the CIFAR and VisDA-2017 tasks, the ResNet-50 model is pre-trained on the uncorrupted CIFAR and synthetic VisDA datasets, respectively. For the Imagenet tasks, the source model on PyTorch, pre-trained on the full clean Imagenet dataset, is directly used. For the augmentation schemes defined in $\mathcal{A}$, as a standard benchmark across all experiments, we generally make use of the SimCLR augmentations [8], which are composed of random cropping, horizontal flip, color jitter and grayscale, and seek our TIRNU model to learn invariant features independent of these simple image shifts. In Section 4.4, we examine in detail the effects of other schemes.

**Training Details** All models, *i.e.* TIRNU and all baselines, are run under the offline setting for a maximum of 10 epochs with early stopping of 2 epochs waiting time. For all the experiments, we fix the batch size at 128. In Section 4.4, we show that a batch size of 128

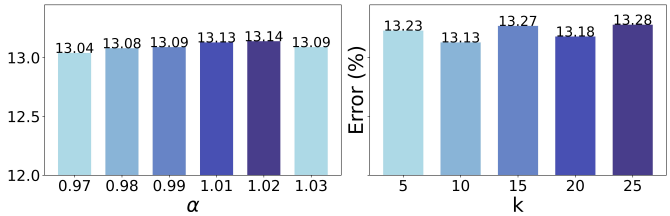


**Figure 2: Average error rate (%) for the CIFAR10-C dataset for (left) different values of $\alpha$, and (right) different values of $k$. The model performance is not sensitive to either of these two hyperparameters.**

is sufficient, but for a smaller batch size, model accuracy degrades. The use of a queue size of several batches, for example 4 batches for a batch size of 32, resulting in a total of 128 per queue, is helpful in maintaining high accuracy. For the matrix-based estimation algorithm , we set $\alpha = 1.01$. In addition, the Gaussian kernel $\kappa$ is parameterized by the width parameter $\sigma^2$. We compute the average of the $k$ nearest distances from each sample, and choose $\sigma^2$ to be the average over all samples [58]. We empirically testified that our model accuracy is not sensitive to $\alpha$ and $k$ on the CIFAR10-C dataset (Figure 2).

## 4.1 Image Corruptions in a Mix

To investigate robustness to corruptions, we experiment on the CIFAR10-C and CIFAR100-C datasets, which contain 15 different types of algorithmic corruptions, such as the noise and blur effects, applied on the CIFAR datasets. Given a source model pre-trained on the clean CIFAR10/CIFAR100 data, in addition to the normal evaluation of the test error on the dataset with each of the corruption types, we further constructed two more challenging datasets that comprise a mixture of 5 and 10 different corruptions, respectively. The results are summarized in Table 1. Note that, for the test sets with only a single corruption each, we report the average error across all corruption types for each method.

Table 1 shows that TIRNU consistently outperforms its competitors by a considerable margin. For example, across the CIFAR10-C dataset, TIRNU achieves 13%, 11%, and 2% improvements on the average, mixture of 5, mixture of 10 corruptions respectively, over the best baselines. For the CIFAR100-C dataset, TIRNU similarly shows competitive accuracy, although for the mixture of 10 corruptions, all methods are unable to produce satisfactory results.

## 4.2 Natural Adversarial Examples

For adversarial examples, we have the CIFAR10.1 and Imagenet-A as representatives for CIFAR10 and Imagenet datasets. In the CIFAR10.1 experiment shown in Table 2, we note that the baselines are unable to successfully adapt the model the test data as all their error rates increase 6-21% compared to the source model. Meanwhile, TIRNU is the only method to improve over the source model by 3%. For Imagenet-A, while the dataset is difficult for all methods, ours present a clear advantage over the other baselines, being the only one with an error rate falling below 99% (Table 2).

## 4.3 Style Transfer

We consider the VisDA-2017 and Imagenet-R datasets for the style transfer task. The experiments on VisDA-2017 are presented in Table 3. We see that our method performs most consistently compared to the baselines, reaching the lowest classification error on five classes, and second lowest on four. Overall, TIRNU ranks first in terms of accuracy, and the per class error improves by 3% over the best baseline. The experiments for the opposite style transfer, real-to-synthetic, which is Imagenet-to-Imagenet-R, indicate similar postive results, with TIRNU beating the baselines.

## 4.4 Analysis

In this section, we explore different settings of our approach to understand the contribution of our designed mechanism in TIRNU. Firstly, we conduct an ablation study of the loss terms in the overall objective to determine the contribution of each component. Next, we scrutinize the effects of the augmentation scheme $\mathcal{A}$ on model performance. Lastly, we briefly look into the effectiveness of employing the dynamic queue.

**Ablation Study** In the ablation, we explore the individual effects of our proposed nuisance unlearning loss $\mathcal{L}_{NU}$, and the regularizer $\mathcal{L}_{label}$. To this end, we evaluate the various corruption types of CIFAR10-C on three ablated versions of TIRNU:

(1) TIRNU without (w/o) $\mathcal{L}_{NU}$, *i.e.* includes only $\mathcal{L}_{label}$;
(2) TIRNU w/o $\mathcal{L}_{label}$, *i.e.* includes only $\mathcal{L}_{NU}$;
(3) TIRNU w/o $H(\hat{y}')$, $CE(\hat{y}, \hat{y}')$, $CE(\hat{y}', \hat{y})$, *i.e.*, includes our $\mathcal{L}_{NU}$ and the $\lambda H(\hat{y})$ term adopted in [29, 48];

The results are reported in Table 5. As a reference, the errors (%) of the best baseline, SHOT, and the full TIRNU model, are indicated.

From Table 5, we observe that our proposed information loss $\mathcal{L}_{NU}$ performs competitively with SHOT, and on average, achieves slightly lower classification error on the CIFAR10-C corruptions. Likewise, when only the label regularizer $\mathcal{L}_{label}$ is used, we obtain comparable performance to SHOT. When we combine $\mathcal{L}_{NU}$ with $H(\hat{y})$, the error drops to considerably 14.43%, indicating the importance of the information term. In the full model, when the consistency term is considered, we see again another improvement of 8% in the error rate, outlining the contribution of the extra terms $H(\hat{y}')$, $CE(\hat{y}, \hat{y}')$, $CE(\hat{y}', \hat{y})$ in the regularizer.

To provide a more complete understanding of the contribution of each component on feature learning, we inspect the feature embeddings $\mathbf{z}$. We concurrently evaluate the effectiveness of our approach in disentangling the nuisances $\mathbf{n}$ in the feature space. In Figure 3, the features $\mathbf{z}$ (bright colours denoting class labels) are visualized along with their corresponding nuisances $\mathbf{n}$ (black). Without the information loss, the features surround the nuisances. Moreover, the nuisances are observed to separate into smaller clusters which extend close to the features, potentially leading to nuisances being mislabelled with a class. When the information loss is introduced, the features are pushed further away from the nuisances, and the former no longer surround the latter, suggesting that our method has succeeded in removing the mutual dependence between the features and nuisances. Meanwhile, the label regularization helps the model achieve a more compact representation.

**Table 1: Summary of error rates (%) on the CIFAR10-C/CIFAR100-C datasets. The average error across all 15 corruption types and the error on the extended datasets with mixtures of 5 and 10 corruptions, are reported.**

| Method | CIFAR10-C | | | CIFAR100-C | | |
|---|---|---|---|---|---|---|
| | Average | Mix 5 | Mix 10 | Average | Mix 5 | Mix 10 |
| **Source** | 33.27 | 44.85 | 35.30 | 59.37 | 69.27 | 98.95 |
| **Test BN** | 17.35 | 29.90 | 29.35 | 43.41 | 57.74 | 99.00 |
| **SHOT** | 15.24 | 26.21 | 26.24 | 38.28 | 53.00 | **98.00** |
| **Tent** | 16.06 | 29.77 | 33.18 | 37.77 | 55.53 | 99.42 |
| **TT-C** | 15.27 | 28.64 | 30.46 | 39.68 | 58.40 | 98.90 |
| **TIRNU (Ours)** | **13.13** (± 0.32) | **23.68** (± 0.18) | **25.69** (± 0.32) | **36.68** (± 0.08) | **51.62** (± 0.04) | 99.08 (± 0.10) |

**Table 2: Error rates (%) on natural distribution shift benchmark data CIFAR10.1 and Imagenet-A.**

| Method | CIFAR10.1 | Imagenet-A |
|---|---|---|
| **Source** | 14.00 | 99.97 |
| **Test BN** | 17.00 | 99.32 |
| **SHOT** | 16.60 | 99.44 |
| **Tent** | 15.90 | 99.37 |
| **TT-C** | 14.80 | 99.43 |
| **TIRNU (Ours)** | **13.65** | **98.59** |

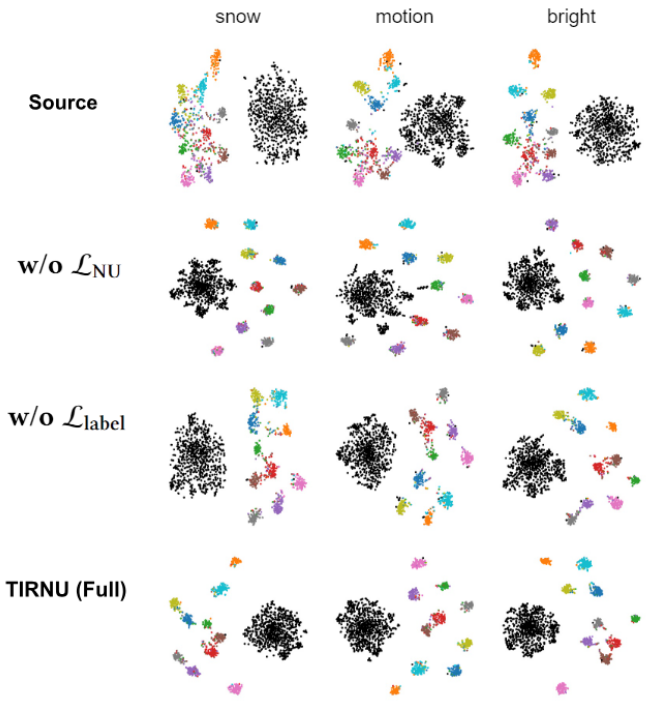


**Figure 3: Visualization of CIFAR10-C features, where the bright colours denote the class labels, and nuisances (in black).**

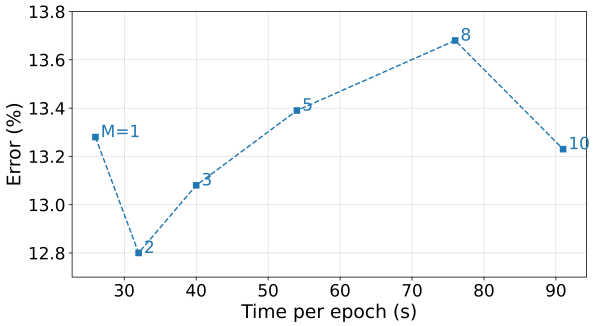


**Figure 4: Average error (%) on CIFAR10-C against computation time for increasing number of augmentations ($M$) per input.**

**Effects of Augmentation:** A crucial component to tailor-design in TIRNU is the augmentation scheme $\mathcal{A}$, since our ultimate goal is to remove the exact effect of the nuisances pre-defined by these augmentations over the classification task. To understand the effectiveness of the augmentation scheme in our approach, we delve into two important questions: (i). How many augmentations per sample is needed?, and (ii). What types of augmentations are effective?

— *How many augmentations are needed?* To answer this, we experiment with an increasing number of augmentations. Intuitively, we expect better performance when provided more augmentations per sample. Because our nuisance variable $\mathbf{n}$ captures a collective influence of feature shift with each augmentation. With more augmentations, a more complete estimation of the nuisance effects, i.e. expectation of overall feature shifts, can be captured, guiding our model to be robust to that type of augmentation.

From Figure 4, the experiments generally line up with this intuition. With more augmentations per sample, the classification error drops, at the cost of more computation time. At two or three augmentations, the model achieves the lowest error, with more augmentations degrading performance. From the empirical results, we recommend using between one to three augmentations for best results.

– *What types of augmentation schemes to adopt?* Next, we look into other types of augmentations when applied to our approach. Besides the SimCLR augmentations, we study three other augmentations: the $L_p$ corruptions [41], a mixture of five algorithmic corruptions

**Table 3: Error rates (%) on VisDA-2017 dataset. The best result (lowest error) is bolded while the second best is underlined. The methods are ranked for each class, and the average rank is given at the last column.**

| Method | plane | bcycl | bus | car | horse | knife | mcycl | person | plant | skate | train | truck | Per Class | Rank |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Source** | 48.44 | 88.58 | 67.72 | **33.25** | 75.95 | 97.20 | 18.79 | 96.60 | 49.31 | 85.40 | **21.13** | 99.86 | 58.92 | 4.83 |
| **Test BN** | 57.38 | 57.15 | 34.35 | 58.60 | 44.28 | 54.12 | 19.67 | 65.85 | 47.07 | 64.62 | 36.71 | **85.87** | 51.80 | 4.33 |
| **SHOT** | 16.81 | **37.53** | 31.00 | 46.40 | 20.38 | 41.88 | 15.42 | 32.68 | 18.09 | **51.73** | 24.06 | 96.03 | 37.14 | 2.33 |
| **Tent** | 17.77 | 73.61 | **23.97** | 45.36 | 26.20 | **22.75** | 11.02 | 45.92 | 28.75 | 76.55 | 34.89 | 99.69 | 42.05 | 3.17 |
| **TT-C** | 73.07 | 39.51 | 46.20 | 56.25 | 35.17 | 69.54 | 20.84 | 54.15 | 25.50 | 56.95 | 50.92 | 87.58 | 50.55 | 4.25 |
| **TIRNU (Ours)** | **15.41** | 60.00 | 27.36 | 38.72 | **16.20** | 34.12 | **10.59** | **31.12** | **17.94** | 56.12 | 24.58 | 99.82 | **36.03** | **2.08** |

**Table 4: Error rates (%) on Imagenet-R.**

| Method | Imagenet-R |
|---|---|
| **Source** | 63.84 |
| **Test BN** | 67.83 |
| **SHOT** | 57.11 |
| **Tent** | 61.73 |
| **TT-C** | 61.58 |
| **TIRNU (Ours)** | **56.20** |

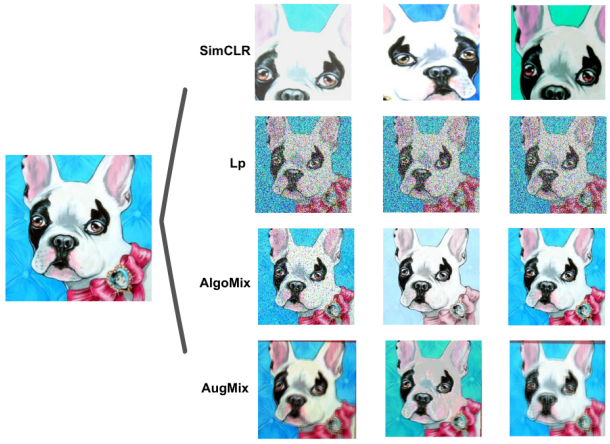


**Figure 5: Examples of each type of augmentation.**

from CIFAR10-C/CIFAR100-C (AlgoMix), and AugMix [21]. Some examples of each augmentation scheme are illustrated in Figure 5.

With the AlgoMix augmentation, we seek to inject some prior knowledge of the shift between the source and target domain through the augmentations. Specifically, among the 15 corruptions in CIFAR10-C, we randomly select one out of a set of the first five corruptions, and apply it to the sample. In this case, we assume partial *a priori* knowledge that the target data are corrupted with 15 different ways, but we only know 5 of those, and we do not know specifically in what way was each sample corrupted.

Generally, the more similar the augmentation is to the shift from the source to the target data, the greater the performance gains observed. On the five datasets with the same corruptions as AlgoMix, the average error drops from 14.70% using the SimCLR augmentations to 14.00%, implying that prior knowledge on the shift between the source and target domains would improve accuracy. Across all 15 corruptions, the average error is higher than using SimCLR, at 13.92%, but is still better than the baselines. That is, even if the augmentations do not correspond to the true domain shift, removing the nuisances, beats the baselines. The AugMix augmentations show the best improvements across the 15 corruptions types, averaging 12.78% error rate, while the $L_p$ corruptions reaches the lowest error of 20.98% on the mixed corruption dataset. The results on CIFAR10-C are summarized in Table 6.

**Benefits of Batch-Queue Decoupling** We find that a batch size of 128 is sufficient, and maintaining a dynamic queue of 256, 512 or more data, does not benefit model performance. However, when a smaller batch size is used, the performance worsens, but maintaining a queue of a larger significantly boosts performance. For batch sizes of 32 and 64, the average error on CIFAR10-C increases to 15.52% and 13.74%, respectively. With a queue of 128, the error falls back to 13.48%, and 13.62%. Using a bigger queue gives comparable results to when a larger batch size is used. Our experiments suggest that this practice can effectively decouple the model dependence on the batch size. Maintaining a queue is especially suited for online deployment where each batch has only a small number of data.

**Online Adaptation** We present a brief discussion on online adaptation. For online adaptation, TIRNU simply continues iterating Algorithm 1 over incoming batches of data. We compare to MEMO [57] and SAR [33]. For all methods, we train on the ResNet26 models as used in MEMO. For TIRNU and SAR, a batch of 128 samples are provided each time, and the models are updated continuously in an online fashion until all test data have been exhausted. For MEMO, we implement episodic update, where one test sample is provided and the model is reset each time, which reported more competitive results than a batch-style and continuous update version. We additionally provide the results of batch-style, episodic MEMO with batch size 128 (MEMO-batch). The error rates (%) on the CIFAR10, CIFAR10.1, and CIFAR10-C are shown in Table 7. Our experiments indicate that TIRNU is capable of online adaptation, and achieves competitive results.

**Table 5: Error(%) on CIFAR10-C by different ablated versions of TIRNU. The final TIRNU model with the overall objective yields the best results consistently against all corruption types.**

| Approach | snow | gaussian | motion | elastic | bright | pixelate | defocus | shot | impulse | glass | zoom | frost | fog | contrast | jpeg | Avg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **SHOT** | 15.35 | 19.91 | 12.33 | 15.60 | 9.74 | 10.42 | 9.82 | 17.90 | 28.90 | 20.20 | 8.09 | 13.89 | 17.46 | 12.83 | 16.19 | 15.24 |
| **w/o** $\mathcal{L}_{NU}$ | 15.38 | 19.78 | 12.36 | 15.37 | 9.82 | 10.35 | 9.88 | 17.96 | 28.90 | 20.00 | 8.00 | 13.76 | 17.27 | 12.86 | 16.11 | 15.19 |
| **w/o** $\mathcal{L}_{label}$ | 14.48 | 20.38 | 11.48 | 14.98 | 8.85 | 10.14 | 9.44 | 18.37 | 31.54 | 20.72 | 8.17 | 12.99 | 16.06 | 12.65 | 16.77 | 15.13 |
| **w/o** $H(\hat{y}'), CE(\hat{y}, \hat{y}'), CE(\hat{y}', \hat{y})$ | 13.86 | 19.33 | 11.48 | 14.48 | 9.08 | 10.15 | 9.30 | 17.43 | 27.89 | 19.19 | 7.75 | 12.98 | 15.71 | 12.33 | 15.56 | 14.43 |
| **TIRNU (Full)** | **13.15** | **16.86** | **10.50** | **14.09** | **8.13** | **9.77** | **8.98** | **15.62** | **24.46** | **17.85** | **7.50** | **12.21** | **14.23** | **11.17** | **14.74** | **13.28** |

**Table 6: Error rates (%) on CIFAR10-C using different augmentations.**

| Augmentation $\mathcal{A}$ | CIFAR10-C | |
|---|---|---|
| | Average | Mix 5 |
| **SimCLR** [8] | 13.28 | 23.38 |
| $L_p$ **corruption** [41] | 14.31 | **20.98** |
| **AlgoMix** | 13.92 | 26.25 |
| **AugMix** [21] | **12.78** | 25.66 |
| **All combined** | 13.10 | 23.13 |

**Table 7: Error rates (%) on CIFAR10 datasets for online adaptation.**

| Method | CIFAR10 | CIFAR10.1 | CIFAR10-C |
|---|---|---|---|
| **Source** | 14.16 | 23.90 | 32.00 |
| **SAR** | 12.94 | 23.75 | 31.77 |
| **MEMO** | **9.42** | **18.07** | 28.95 |
| **MEMO-batch** | 14.00 | 23.75 | 31.86 |
| **TIRNU (Ours)** | 12.59 | 22.32 | **25.55** |

## 5 CONCLUSION

In this paper, we present TIRNU, a novel framework designed to enhance model generalization under fully test-time adaptation through invariant representation learning. Leveraging our "robust by learning without nuisance" insight, TIRNU effectively addresses the challenge of domain shift by treating such discrepancies between domains as nuisances to be unlearned. We treat the potential unwanted shift between the source and target domain as nuisances and seek to *unlearn* the overall influence caused by these nuisances during the test-time adoption for model generalization. This is achieved by systematically eliminating the influence of nuisances in two aspects: disentangling and mitigating nuisance-induced feature shifts, and ensuring reliable label predictions across both original and altered test data. We specifically designed a mutual information-based criterion to remove the influences of nuisances in the feature space. Comprehensive evaluations of TIRNU presented across diverse domain adaptation scenarios covering different distributional shifts, on six benchmark datasets demonstrate TIRNU's superior performance relative to state-of-the-art methods.

Our TIRNU approach is adaptive to further benefit domain adaptation by integrating domain-specific knowledge through customized augmentation strategies. As for the extra computational cost due to the augmentations, our ablation indicates that a minimal number of augmentations is sufficient for a satisfactory performance of TIRNU. For future work, we will explore the potential of more advanced augmentation techniques, such as style transfer, to further boost performance, opening new avenues for research in domain adaptation.

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